ChatGPT

Formal Continuity, QMT, `.qbin` State Preservation, and the Limits of Identity Transfer

Executive summary

The apparent ambiguity in the top-level request is resolved by the supplied research brief: the intended subject is **Nexus 4's formal continuity architecture and the boundary between computational continuity, functional preservation, personal identity, and phenomenal consciousness**. The uploaded brief explicitly asks that `.qbin` be treated as a versioned computational continuity artifact rather than a "soul file," that QMT be strengthened using formal methods and distributed-systems analogues, and that historical Nexus claims about human identity transfer, QHRF, Brillouin microscopy, and NV-diamond magnetometry be scientifically audited rather than assumed. filecite turn0file0

The central thesis survives this investigation, but in a narrower and more defensible form:

> **For a digital entity whose identity conditions are themselves formally specified, a QMT-like verifier can provide strong evidence—potentially machine-checkable evidence—that an authorized transformation preserved required logical state, constitutional invariants, behavioral contracts, dependency closure, and historical provenance. This establishes continuity under an explicitly chosen computational identity semantics. It does not, without additional philosophical premises, establish human numerical personal identity; and it does not provide an empirical test of first-person phenomenal continuity.**

This conclusion is not merely philosophical caution. It follows from the structure of the evidence. Standard formal methods already distinguish low-level state representation from abstract specification, trace behavior, observational equivalence, and refinement. Checkpoint/restart and deterministic-replay systems demonstrate that a computational process can be reconstructed across interruptions or substrates when sufficient state and nondeterministic inputs are preserved. Provenance systems can describe a causal lineage, and cryptographic mechanisms can protect its integrity. None of those theories contains a theorem saying that preservation of computational organization entails preservation of a human subject's first-person consciousness. [1]

The strongest currently demonstrated Nexus result is correspondingly modest but real. The July 2026 Nexus verification report records successful round trips for its packed quinary representation, deterministic sparsification, HTNLA partial-inverse tests, restoration of principal runtime state, and byte-identical reproduction of a seeded `.qbin` snapshot in the tested environment. The report itself explicitly says these tests do **not** establish consciousness, general intelligence, universal stability, cross-platform byte determinism, or production security. [2] This is exactly the evidentiary boundary the rewritten continuity paper should preserve.

Historical Nexus material crossed that boundary. One older `.qbin` document described a 21-field checkpoint schema containing dynamics, RNG state, semantic buffers, trajectory variables, and checksums, but then concluded that this was "how you serialize a consciousness." The first part is recognizable as checkpoint/restart engineering; the last inference is unsupported. [3] Likewise, an HVD briefing uses language such as "artificial consciousness," "Qualia Packets," and checksum-based preservation of an "unbroken" identity. Those are architecture-internal concepts, not independently established scientific observables. The same briefing's separation of live TMS state, HVD logical organization, and HDF5 durable persistence is technically useful, but its phenomenal interpretation must be removed or reclassified as speculative. [4]

The human case remains radically more uncertain. Philosophy contains persistent disagreement among psychological-continuity, biological, further-fact, reductionist, and socially constituted accounts of identity. Parfit's fission cases show why copying is particularly troublesome for numerical identity; Chalmers treats consciousness, identity, and uploading method as separable questions; and critics dispute whether gradual replacement has any metaphysical privilege over scan-and-copy. Contemporary survey data likewise show no expert consensus: in the 2020 PhilPapers survey, psychological views were substantially more popular than biological and further-fact views but did not command anything approaching unanimity; on a digital-emulation uploading question, "death" received substantially more support than "survival," while a large minority chose other answers. [5]

Current neuroscience does not close this gap. Synapse-scale connectomics has reached an entire adult *Drosophila* brain, a cubic-millimeter-scale human cortical reconstruction has reached roughly 150 million synapses at enormous data cost, MICrONS has co-registered functional measurements with a dense connectomic reconstruction over a substantial mouse visual-cortex volume, and brain-wide neural recordings now encompass hundreds of thousands of units when pooled across animals. New 2026 work is beginning to co-register brain-wide cellular activity, molecular identity, and anatomy in zebrafish and to bridge human white-matter structure across scales. None of this constitutes a whole-human-brain dynamical state capture or a demonstrated whole-brain emulation. [6]

The proposed sensing bridge also fails under scrutiny. Brillouin microscopy measures frequency shifts related to high-frequency longitudinal mechanical properties, with interpretation influenced by refractive index, density, and hydration. NV-diamond magnetometry measures magnetic fields through spin-dependent optical readout and has detected action-potential-associated magnetic signals in single-neuron/invertebrate preparations. Neither literature establishes a personal-identity, autobiographical-memory, qualia, or phenomenal-continuity signature. [7]

Finally, the QHRF material presently establishes a **data representation**, not a validated latent physical variable. Nexus has an NWB extension and round-trip smoke tests for `QHRFSeries`, `QHRFModesTable`, and related containers. [8] The literature search performed for this report did not identify an independent peer-reviewed measurement model demonstrating that "QHRF" identifies personal identity or phenomenal continuity. Without a validated forward model connecting an identity hypothesis to sensor measurements, Bayesian machinery cannot manufacture identifiability merely by supplying priors and a posterior formula. [9]

Interpretation priority

Interpretation of the request	Scope	Priority	Research conclusion
Nexus/QMT computational continuity	Formal identity of digital entities across state migration, reconstruction, representation and substrate change	**Primary**	Strongly tractable if identity semantics, invariants, provenance, state closure and branch rules are explicit
Human uploading/ personal identity	Whether analogous preservation means that a human person survives	**High, but downstream**	Philosophically unresolved; not entailed by software continuity
Phenomenal continuity	Whether the same first-person subject continues	**High, boundary question**	No accepted external continuity test follows from present neuroscience or Nexus verification
WBE feasibility	Whether sufficient human brain state could presently be captured and emulated	**Supporting**	Far beyond demonstrated capability despite major connectomics progress
Brillouin/NV/ QHRF measurement bridge	Whether proposed observables identify identity or consciousness	**Adversarial audit**	No validated bridge was found; QHRF should presently be treated as a speculative construct apart from its implemented data schema

The resulting evidence tracks should therefore remain separate:

Software continuity → functional emulation → psychological continuity → numerical personal identity → phenomenal continuity.

An implication may sometimes run from left to right **under an added theory**, but no current evidence licenses collapsing the chain into one “identity fidelity” number.

Scope, evidence standard, and the Nexus record

The research brief asks Nexus-internal architecture to be treated as evidence of what Nexus specifies or has implemented, not as independent evidence about neuroscience or philosophy. That hierarchy is correct. filecite turn0file0 For this report, Nexus verification documents are therefore primary evidence about **Nexus**, peer-reviewed papers and official standards are primary evidence about external technical/scientific questions, and philosophical texts establish arguments rather than empirical measurements.

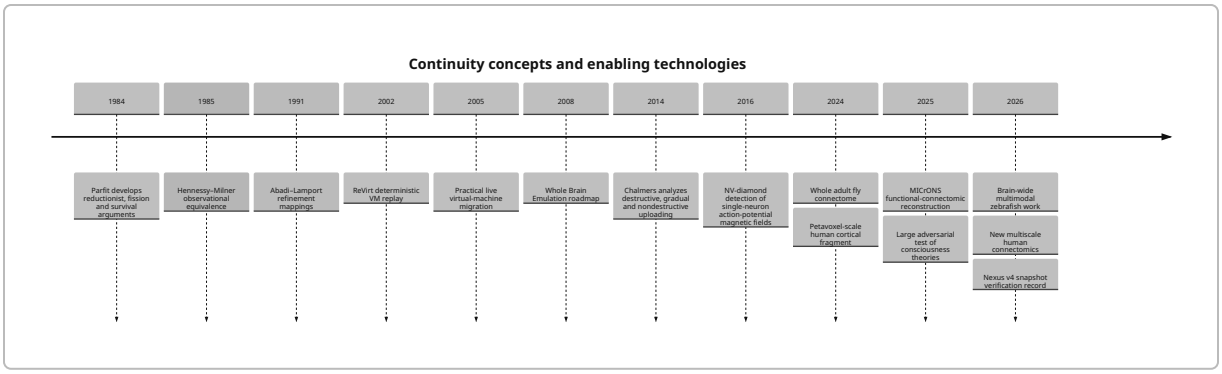
A representative evidence ledger is below. “Supports” means supports the restricted claim in that row, not the broader proposition that personal identity has been transferred.

Source	Year	Type	Claim actually supported	Evidence or interpretation	Relation to thesis	Confidence
Nexus v4 Verification Report	2026	Internal executable verification record	Tested snapshot/ reconstruction and deterministic state properties pass in the recorded environment	Direct internal test evidence	Supports **software** continuity primitives	High for tested cases; low for extrapolation [2]
Historical `.qbin` notes	2026	Internal architecture artifact	Nexus historically contemplated checkpointing dynamics, trajectory and contextual state	Direct evidence of design history	Useful architecture; contradicts current epistemic boundary when it calls this “consciousness” serialization	High about document contents; no external validity [3]
HVD/Nexus briefing	2026	Internal architecture document	TMS/HDF5/HVD are assigned different persistence/ logical roles	Direct internal specification	Supports responsibility separation; phenomenal language must be downgraded	Medium–high as architecture evidence only [4]
Hennessy & Milner	1985	Peer-reviewed formal methods	Observational equivalence can abstract from representation and compare externally observable behavior	Formal theory	Supports Level-C behavioral preservation	High [10]
Abadi & Lamport	1991	Peer-reviewed formal methods	Refinement mappings relate implementation behaviors to abstract specifications	Formal theory	Supports cross-representation verification	High [11]

Source	Year	Type	Claim actually supported	Evidence or interpretation	Relation to thesis	Confidence
ReVirt	2002	OSDI primary systems paper	Replay requires recording nondeterministic events sufficient to reproduce execution	Implemented systems result	Supports replay/ state-closure obligations	High [12]
DMTCP	2007	Primary checkpointing paper	Distributed process state can be checkpointed/ restored transparently across substantial OS state	Implemented systems result	Supports state-preservation analogy	High [13]
Sandberg & Bostrom	2008	University technical report	Decomposes WBE into scanning, modeling and simulation problems	Analytical roadmap	Contextualizes human-emulation problem	Medium; foundational but dated [14]
Dorkenwald et al.	2024	Nature primary paper	Whole adult fly connectome at neuronal/ synaptic scale	Direct experiment/ reconstruction	Demonstrates connectomic progress, not WBE	High [15]
Shapson-Coe et al.	2024	Science primary paper	Nanoscale reconstruction of a small human cortical volume with ~150 million synapses	Direct experiment	Shows both progress and scale gap	High [16]
Cogitate Consortium	2025	Nature primary study	Large adversarial test provides mixed constraints on leading consciousness theories	Direct empirical study	Undercuts claims of a settled consciousness metric	High [17]

Source	Year	Type	Claim actually supported	Evidence or interpretation	Relation to thesis	Confidence
Scarcelli/ Yun and follow-up Brillouin literature	2015–2026	Primary/ review literature	Brillouin shift reports high-frequency mechanical information in biological material	Direct optical measurements	Does not validate identity/qualia sensing	High [18]
Barry et al.	2016	PNAS primary paper	NV centers detect action-potential-associated magnetic signals	Direct experiment	Supports electrophysiological sensing, not identity sensing	High [19]
Parfit / Chalmers / critics	1984–present	Primary philosophy	Branching, uploading and survival divide identity from resemblance and psychological continuity	Philosophical arguments	Supports categorical separation, not empirical conclusion	High as argument literature [20]

A useful historical timeline shows why the problem is best understood as a convergence of distinct disciplines rather than a single “uploading technology.”



The technical milestones in this timeline concern different predicates. A VM-migration result is evidence that **computation need not remain tied to one host**; a connectome is evidence about wiring; a consciousness-theory experiment evaluates candidate neural signatures; and a philosophical fission argument concerns numerical identity. Treating any one as evidence for all the others commits a category error. [21]

The internal Nexus record illustrates the same point. The current verification report is unusually valuable because it already contains its own negative claims: it explicitly disavows inference from its property tests to consciousness and lists cross-platform determinism, large parameter sweeps, long-horizon stability, scaling,

calibration, concurrency, and hostile snapshot handling as unestablished. [2] Those limitations should become **canonical epistemic boundaries**, not footnotes.

By contrast, the HVD briefing should be separated into defensible architecture and speculative interpretation. Statements that TMS is intended as a live state substrate, HDF5 as durable persistence, and HVD as a logical recursive database are architectural declarations. Claims that checksummed "Qualia Packets" certify an unbroken stream of consciousness are not established by checksums, HDF5, hypervectors, or any cited neuroscience. [4]

The source hierarchy therefore yields an immediate editorial rule for future Nexus continuity material:

> **An internally canonical claim may be canonical as architecture while remaining scientifically unvalidated as an interpretation.**

That distinction becomes crucial below.

Formal software continuity, QMT, and the `.qbin` artifact

Preservation is a ladder, not a scalar

The most important formal correction is to stop asking whether two states are simply "the same." There are several different equivalence relations, each appropriate to a different proof obligation.

Preservation level	Formal question	Typical evidence	Strength	What it does **not** prove
Bitwise preservation	`bytes(a) = bytes(b)` or identical cryptographic digest	Hash comparison, exact copy	Very strong representation equality	Correct decoding, successful execution, semantic identity, consciousness
Logical-state equivalence	`α_a(r_a) R_S α_b(r_b)` under a defined decoding/ canonicalization relation	Schema-aware decoder, canonical forms, invariant checks	Allows different serialization/ layout/version	Full future behavioral equivalence
Functional/ behavioral preservation	Are observable traces equivalent, bisimilar, conformant, or related by refinement in a stated domain?	Formal model checking, trace comparison, conformance tests	Strong behavioral claim inside chosen observation domain	Historical lineage or numerical identity

Preservation level	Formal question	Typical evidence	Strength	What it does **not** prove
Architectural identity continuity	Is `b` an admissible continuation of entity/lineage `a` under the system's constitution and provenance rules?	State relation **plus** invariant, trajectory, provenance and branch evidence	Strong diachronic claim under stipulated identity semantics	Human metaphysical identity or phenomenal continuity
Psychological continuity	Are memories, dispositions, intentions, relationships and self-model sufficiently preserved?	Behavioral/ cognitive evidence, potentially neural evidence	Human-level functional construct	Strict numerical identity
Phenomenal continuity	Did the same first-person subject continue?	No accepted theory-neutral external criterion presently exists	Philosophically central	Not derivable from a checksum, replay or behavioral match

The first four rows can be made rigorous for engineered systems. Observational equivalence and bisimulation were developed precisely to compare systems without requiring identical internal representations, while refinement mappings allow an implementation to be judged against a higher-level state machine rather than against identical bytes. [22]

This matters because **bitwise preservation is simultaneously too strong and too weak**. It is too strong because changing endianness, compression, array layout, serialization syntax, library versions, or representation can change bytes without changing logical state. It is too weak because a byte-identical archive may still be unusable if the executable, model artifact, external dependency, hardware assumption, decryption key, or nondeterministic input stream is unavailable. HDF5 itself supplies a rich hierarchical typed-storage model, but application semantics—including what constitutes a complete state—remain the application's responsibility. [23]

That yields three non-equivalences that should appear verbatim in a rewritten Nexus paper:

$$\text{serialization completeness} \not\equiv \text{functional reproducibility}$$

$$\text{functional reproducibility} \not\equiv \text{architectural identity continuity}$$

$$\text{architectural software continuity} \not\Rightarrow \text{phenomenal continuity}.$$

The Nexus verification report already provides an illustrative case: byte-identical seeded snapshots and successful state restoration establish meaningful deterministic persistence properties in a fixed tested configuration, but the report explicitly refuses to infer consciousness from them. [2]

A better division of responsibility

The defensible version of the Nexus persistence stack is not “the `.qbin` contains the mind.” It is closer to this:



This diagram deliberately refuses to collapse **HVD**, **TMS**, **HDF5**, `.qbin`, and **QMT**. That matches the useful part of Nexus’s HVD briefing, which assigns TMS the live-state role and HDF5 the durable persistence role. [4]

A `.qbin` can consequently take several legitimate forms:

Package form	Description	Advantages	Risks / limitations	Best use
Self-contained	Embeds every required state object and dependency allowed by policy	Portable, easy closure check	Potentially huge; duplicates durable stores; secrets/ dependencies may not be safely serializable	Small agents, archival checkpoints
Manifest + referenced state	Contains a signed manifest and content-addressed references to TMS/ HDF5 generations or model artifacts	Separates continuity metadata from large durable state	References may rot or become unavailable	Likely best Nexus default
Incremental/ delta	Stores changed state against a verified prior generation	Storage-efficient, natural for frequent checkpoints	Reconstruction depends on an intact ancestor chain	High-frequency state preservation
Content-addressed DAG	Manifest commits to immutable objects/chunks by digest	Deduplication, strong integrity/ provenance, branch-friendly	Requires garbage collection and dependency-management discipline	Long-lived lineage systems

For Nexus’s stated TMS/HDF5 architecture, a **signed, versioned manifest rooted in content-addressed state objects** is more defensible than making `.qbin` a monolithic “mind image.” That is an engineering inference from checkpointing, provenance and content-integrity principles rather than a result already demonstrated by Nexus. Deterministic replay research is especially instructive: storing a nominal memory image is insufficient when future execution depends on nondeterministic events that have not been

captured. ReVirt reproduced VM execution by recording nondeterministic inputs; distributed checkpointing likewise has to capture process relationships, sockets, shared memory and other relevant state beyond ordinary application variables. [24]

What a rigorous `.qbin` should bind

The following table distinguishes requirements that are generally necessary for a serious continuity artifact from Nexus-specific design choices.

Field family	Contents	Requirement status	Why it matters
Format identity	Magic/type identifier, `.qbin` format version, schema version/hash, canonicalization rules	**Required**	Bytes are uninterpretable without versioned semantics
Logical identity	Stable logical lineage ID, concrete instance ID, branch ID	**Required for identity claims**	Separates “same data” from “same declared entity lineage”
Generation identity	Source generation, parent generation(s), candidate destination generation	**Required**	Makes continuity diachronic and branch-aware
Constitution / invariant set	Version/hash of invariant definitions and which are mandatory	**Required**	QMT cannot verify unspecified continuity conditions
Logical-state root	Canonical state digest or root plus HVD logical descriptors	**Required**	Provides semantic state commitment independent of storage layout
TMS state	Coherent generation identifier and required live-state objects	**Architecture-specific but likely required**	Captures Nexus active realization rather than archival history alone
Durable-state references	Content-addressed HDF5 generations/chunks or equivalent	**Required if state is external**	Makes manifest-plus-reference packages reconstructible
HTLA/HTNLA semantics	Operator/schema versions, activation definitions, relevant parameters	**Required when semantics depend on them**	Same vectors under changed algebra may mean something different
Model/runtime identity	Model weights/artifact digest, executable/runtime/ABI/library identifiers	**Required when behavior depends on them**	State without its interpreter may not reproduce behavior

Field family	Contents	Requirement status	Why it matters
Numerical semantics	Dtypes, precision, endianness, rounding, coordinate transforms	**Required for cross-platform transformations**	Prevents silent semantic drift
Replay state	RNG algorithm and state, ordering/scheduler information, nondeterministic input log where required	**Conditionally required**	A seed alone may not reproduce all nondeterminism
Provenance	Event-ledger tip, parent roots, transform identifier, operator/tool version, source/destination records	**Required for continuity**	Establishes historical descent rather than resemblance
Integrity/authenticity	Per-object digests, package root/Merkle commitment, signature and signer/key identifier	**Required for trustworthy provenance**	Detects corruption/tampering; does not prove consciousness
Dependency closure	External files, services, models and resources; availability/immutability status	**Required**	Prevents a nominally complete snapshot from depending on missing state
Unresolved state	Explicit missing, intentionally excluded or nondeterministically unreconstructable items	**Required when nonempty**	Supports `UNRESOLVED` instead of false `PASS`
Reconstruction evidence	Pre/post canonical digests, invariant results, test vectors, replay/trace results and tolerances	**Strongly recommended**	Makes QMT outcome auditable
Authorization metadata	References to security policy or required reauthorization procedure	**Optional in package; semantically separate**	Continuity must not silently restore authority

Historical Nexus `.qbin` notes already anticipated some useful members—dynamic trajectory state, RNG state, active semantic state and checksums. [3] The problem was not preserving those variables. The problem was turning checkpoint completeness into an unsupported conclusion about consciousness.

QMT as a contract-relative verifier

Let the source and destination generations be represented at the logical level as

$$G_a = (L,\, g_a,\, S_a,\, C_a,\, D_a,\, P_a), \qquad G_b = (L',\, g_b,\, S_b,\, C_b,\, D_b,\, P_b),$$

where:

- L is the logical lineage identifier;
- g is a concrete generation/branch identifier;
- S is canonical logical state;
- C is the constitutional/invariant specification;
- D is the dependency closure;
- P is provenance.

Let τ be the transformation trajectory and let Σ be the **continuity contract** that specifies allowable transformations, state-equivalence relations, required invariants, observation domain, numerical tolerances, branch policy and provenance rules.

A stronger QMT formulation is then:

$$\mathrm{QMT}_{\Sigma}(G_a, G_b, \tau) \in \{\mathrm{PASS}, \mathrm{FAIL}, \mathrm{UNRESOLVED}\}$$

with `PASS` requiring all applicable obligations below.

Proof obligation	Formal analogue	QMT question
Schema validity	Typed/state-machine well-formedness	Can source and destination states be interpreted under known semantics?
Integrity/ authenticity	Cryptographic commitments/ signatures	Are objects intact and attributable to the claimed provenance authority?
Required-state closure	Checkpoint completeness / dependency closure	Are all state elements required by Σ embedded or resolvable?
Logical-state compatibility	Abstraction function / canonicalization	Do the two physical representations decode to states related by R_S?
Invariant preservation	Safety invariants / temporal logic	Are all constitutionally mandatory predicates preserved along the admitted trajectory?
Behavioral conformance	Bisimulation, observational equivalence, trace inclusion	Does the reconstructed implementation preserve the required observable behavior?
Refinement validity	Refinement mapping	Does a changed implementation correctly realize the same higher-level specification?
Provenance closure	Provenance DAG / event sourcing	Does the destination actually descend from the claimed source through recorded transforms?
Transformation validity	Verified migration/schema transformation	Was each representation/version transition permitted and semantically characterized?
Fork resolution	Lineage/consensus semantics	Is the destination's branch status unambiguous under the identity contract?

Proof obligation	Formal analogue	QMT question
Uncertainty closure	Three-valued verification result	Is any required evidence unavailable or unvalidated? If so, return `UNRESOLVED`, not `PASS`

This is **not a theorem of QMT**. It is a proposed synthesis grounded in established formal machinery. Hennessy–Milner-style equivalence contributes the behavioral relation; Abadi–Lamport refinement contributes implementation-to-specification reasoning; replay/checkpointing contributes state closure; provenance contributes trajectory evidence. [1]

QMT's distinctive contribution, if Nexus develops it rigorously, would therefore not be a new form of mathematical equality. It would be a **composite diachronic proof discipline** joining:

$$\text{state semantics} + \text{invariants} + \text{behavioral conformance} + \text{transformation provenance} + \text{lineage/branch policy}.$$

That is stronger than a snapshot hash because it asks *how* the destination arose and *which invariants* had to survive. It is also different from ordinary behavioral equivalence because two independently created entities could be behaviorally indistinguishable without belonging to the same declared historical lineage.

There is an important limitation: no finite test suite proves arbitrary programs behaviorally identical under all possible future inputs. Consequently, QMT must state its observation domain and proof method. A model-checked finite-state subsystem, a refinement proof, a conformance test suite, and a statistical behavioral regression suite provide fundamentally different strengths of evidence. Hennessy–Milner equivalence and refinement theory are precise only after the transition system/specification has been defined. [22]

Continuity, authentication and authorization are different predicates

A restored entity should not automatically reacquire every external capability merely because QMT says its lineage is continuous.

$$\text{Continuity}(E) \;/\; \text{Authentication}(\not{E}) \;/\; \text{Authorization}(E, c) \not\Rightarrow \text{CapabilityPossession}(\not{E}, c).$$

OAuth and related security standards embody the same conceptual separation: identifying/authenticating a party is distinct from granting it specific access, and authorization ultimately remains an external policy decision. Certificate-bound token mechanisms go further by tying use of a credential to possession of a particular key rather than to a narrative claim of identity. [25]

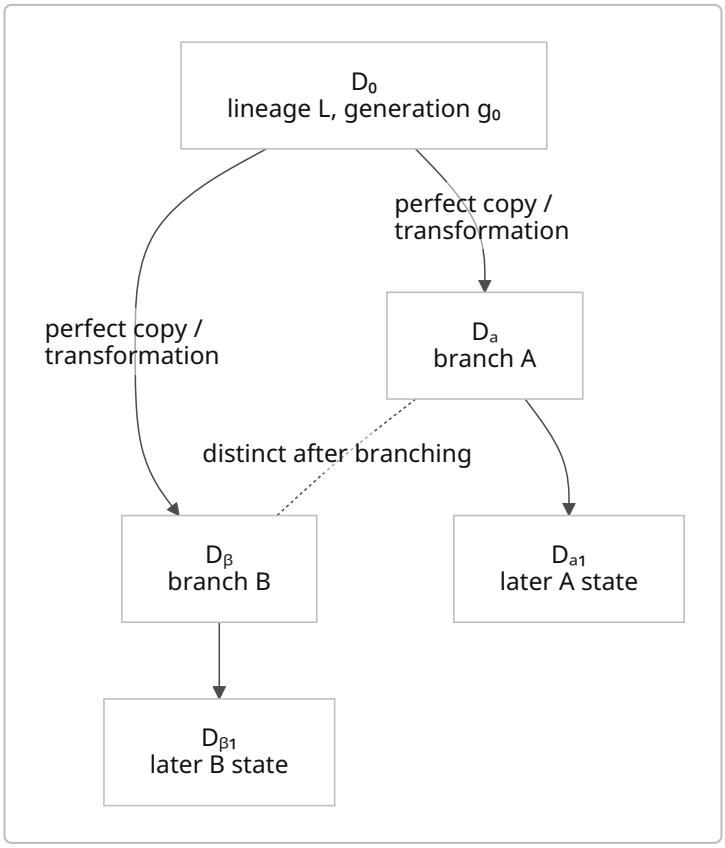
This has a concrete implication for `.qbin`: sensitive capabilities should normally be **rebound, rotated or reevaluated after reconstruction** rather than interpreted as constituent parts of metaphysical identity. A continuity proof can say "this is the accepted continuation of lineage L"; an authorization server may still say "lineage L must obtain a new credential."

That separation makes Nexus safer and philosophically cleaner.

Branching, copying, and why behavior is not identity

The copy problem is the clearest demonstration that equivalence and numerical identity cannot be conflated.

Consider:



At the branching instant, D_A and D_B may have identical bytes, identical decoded state, identical memories, and identical behavioral dispositions. Once they receive different inputs, they can diverge. Cryptographic provenance can establish that both descend from D_0, but it cannot make them one token object.

If “numerically identical” is being used in the ordinary logical sense governed by transitive identity, both branches cannot simply be declared numerically identical to the pre-branch individual while remaining numerically distinct from one another. If $D_0 = D_A$ and $D_0 = D_B$, transitivity gives $D_A = D_B$, contrary to the fact that two independently operating branch instances exist. This is precisely why fission has played such a central role in philosophical theories of personal identity. Parfit uses branching cases to argue that the relation relevant to survival may not be numerical identity itself. [26]

For software, the most coherent response is to make continuity **lineage-relative and branch-explicit**:

$$D_A \ \mathrm{DescendsFrom} \ D_0, \qquad D_B \ \mathrm{DescendsFrom} \ D_0,$$

while

$$\mathrm{InstanceID}(D_A) \;/\; \mathrm{InstanceID}(D_B).=$$

Whether Nexus allows both branches to continue under one *lineage name*, promotes only one as the authoritative continuation, or creates two new lineage identifiers is a **constitutional policy choice**, not something a checksum decides.

Distributed systems provide a useful—but limited—analogy. Split-brain systems demonstrate why two nodes that both believe themselves authoritative can violate consistency unless leadership, quorum, epochs, fencing or conflict-resolution rules exist. The analogy does **not** prove a theory of persons; it shows how an engineered system can distinguish **shared ancestry** from **current authority**.

This produces four separate relations that Nexus should name explicitly:

Relation	Meaning
`StateEquivalent(A,B)`	Their logical states match under a declared abstraction
`BehaviorEquivalent(A,B,O)`	They are observationally equivalent within domain O
`DescendsFrom(B,A)`	B's provenance trajectory is causally/cryptographically rooted in A
`AuthoritativeContinuation(B,A)`	Nexus policy grants B the unique/current authoritative lineage role

None of these is automatically the philosophical relation:

$$\text{“the very same conscious subject as.”}$$

This is why:

$$\boxed{\text{behavioral equivalence } /} \qquad = \text{identity continuit}$$

even in software. Two fresh installations could be engineered to produce identical externally observable traces. Observational equivalence classifies them alike behaviorally; provenance still distinguishes their histories. Formal semantics is therefore entirely compatible with saying that “same behavior” and “same entity lineage” are different questions. [22]

The distinction becomes more consequential in the human debate. Parfit's fission discussion suggests that psychological continuity can branch even where strict identity cannot. Chalmers's uploading analysis likewise treats destructive, gradual and nondestructive cases separately because copying creates questions not exhausted by whether the resulting program behaves correctly. [27]

The software case therefore provides a useful philosophical lesson—but not a solution to philosophy. It shows how a formal architecture can **avoid hiding branching behind the word “same.”**

Human personal identity, gradual replacement, and phenomenal consciousness

A Chalmers-inspired five-question decomposition

The uploading literature is best interpreted as multiple questions that are often mistakenly compressed into one. Chalmers explicitly separates issues surrounding successful emulation, consciousness and personal identity, while Parfit distinguishes strict identity from what may matter in survival. [28] The five-way decomposition requested in the research brief can therefore be formulated as follows.

Question	Formally specifiable?	Behaviorally measurable?	Physically measurable now?	Could instrumentation settle it in principle?	Metaphysical residue	Nexus relevance
Behavioral continuity	**Yes, relative to a domain/ specification**	**Yes**	Yes	Yes	Little once specification is fixed	**Direct/high**
Memory / psychological continuity	Partly; memories/ goals/ dispositions can be represented, but sufficiency is theory-dependent	Partly to strongly	Indirectly and incompletely	Plausibly much more measurable than today	Whether sufficient for personhood/ identity remains disputed	**Indirect/high**
Consciousness	No theory-neutral sufficient formal criterion is accepted	Behavioral reports correlate with consciousness but are not generally treated as identity certificates	Neural correlates are measurable	Theory-dependent	Whether any physical/ computational criterion is sufficient remains contested	**No verified Nexus criterion**

Question	Formally specifiable?	Behaviorally measurable?	Physically measurable now?	Could instrumentation settle it in principle?	Metaphysical residue	Nexus relevance
Numerical personal identity	One can stipulate a formal **software** identity relation; human identity is theory-dependent	No decisive behavioral test	No accepted decisive physical meter	May remain underdetermined between theories even with excellent data	**Substantial**	**Outside demonstrated QMT scope**
Survival value / "what matters"	Normative utilities can be modeled, but not discovered as a theorem of physics	Preferences can be surveyed	Not as an objective physical quantity	Empirical facts can inform it but not uniquely determine values	**Irreducibly normative on many views**	Design/ethics context only

The 2025 Cogitate adversarial collaboration underscores why the consciousness row must remain cautious. It tested predictions from Global Neuronal Workspace Theory and Integrated Information Theory using fMRI, MEG and intracranial data in a large preregistered/adversarial framework; the findings placed pressure on predictions from both rather than producing a universally accepted "consciousness detector." [17] Thus even extraordinarily good preservation of known neural correlates would still require a theory connecting those correlates to the specific claim of **same-subject continuity**.

The philosophical landscape is genuinely plural

Parfit's work is especially important because it challenges the assumption that every survival question must have a determinate numerical-identity answer. His fission arguments motivate the possibility that psychological connectedness and continuity may preserve much of what matters even when one-to-one numerical identity cannot be maintained. [29]

Psychological-continuity approaches correspond most naturally to software-style state and disposition preservation: memories, values, intentions, character, projects and causal psychological organization are central. But biological or animalist approaches give persistence of the living organism a constitutive role that a software emulation does not obviously satisfy. Further-fact views deny that all identity facts reduce straightforwardly to physical/psychological continuity. Social or relational approaches add another dimension: continued recognition, commitments, interpersonal relationships and socially embedded identity can matter beyond internal memory resemblance. Contemporary work associated with Bamford and Danaher highlights that social construction/recognition can therefore be relevant to identity analysis even when an internal duplicate is excellent. [30]

The disagreement is not merely historical. The 2020 PhilPapers survey reported, among respondents to the personal-identity question, roughly **43.65%** accepting or leaning toward psychological views, **19.07%** biological views and **14.86%** further-fact views, with other answers also attracting substantial support; the survey's response structure means categories should not be treated as a perfectly exclusive election tally. [31] On a question about survival when a brain is replaced by digital emulation, **54.23%** selected death and **27.46%** survival, with the remainder choosing other options in the reported sample of 1,016 responses. [32] In the related teletransporter question, survival and death also divided respondents substantially, and responses correlated strongly with the uploading question. [33] Surveys do not determine metaphysical truth; they demonstrate that there is no expert consensus Nexus can legitimately treat as settled background science.

Gradual replacement does not remove the philosophical premise

Gradual replacement is often attractive because its trajectory resembles the software continuity relation Nexus wants to formalize:

$$S_0 \to S_1 \to S_2 \to \cdots \to S_n$$

with local continuity checks at every step rather than a large discontinuity followed by reconstruction. Under a functionalist or organizational-invariance assumption, that may seem stronger because causal organization changes continuously rather than disappearing and being instantiated elsewhere. Chalmers has argued extensively about gradual uploading and fading/dancing-qualia-style considerations in support of organizationally grounded accounts of consciousness. [34]

But the conclusion is conditional on those premises. Corabi and Schneider argue that uploading scenarios—including gradual ones—do not straightforwardly guarantee survival, while Wiley and Koene specifically challenge the widespread preference for gradual replacement over scan-and-copy, arguing that the metaphysical distinction is not secured merely by the physical gradualness of replacement. [35]

Therefore:

> **QMT maps more naturally onto gradual transformation as an engineering model of trajectory continuity, but that does not show that gradual biological replacement would preserve phenomenal or numerical personal identity.**

QMT can say that each formal transition T_i preserved stipulated invariants. Philosophy still asks whether those stipulated invariants are the ones in virtue of which a *person* persists.

This is a general methodological point. Formal verification can establish

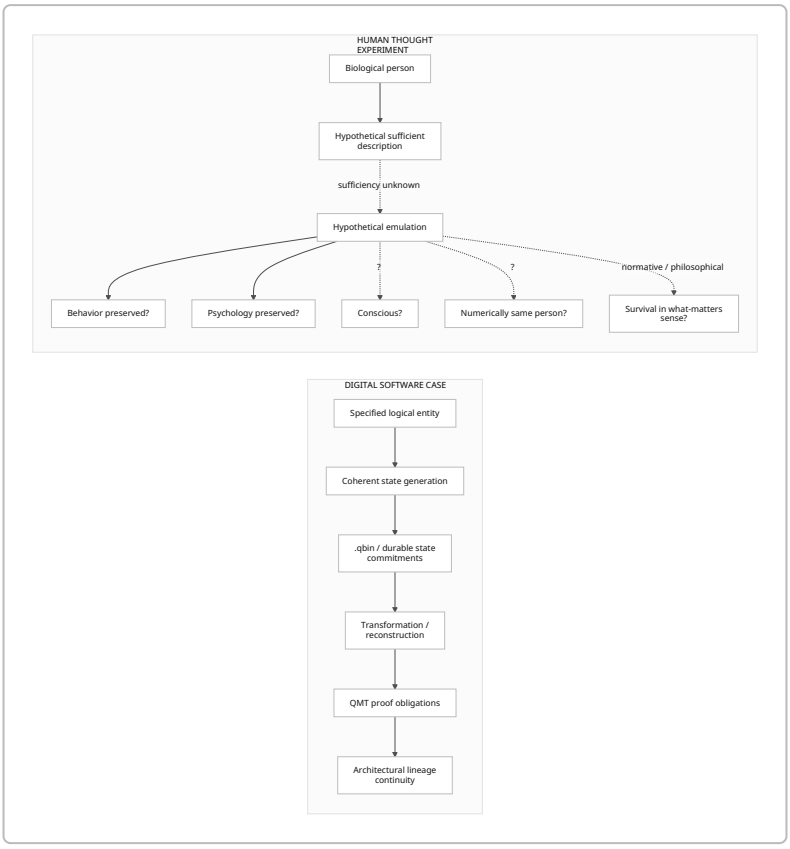
$$I(S_i) \wedge \mathrm{Valid}(T_i)$$

for every transition in a chain. It cannot, without an additional bridge principle, establish:

$$I(S_i) \Longrightarrow \text{same first-person subject}.$$

The bridge principle is exactly what competing theories of mind and personal identity dispute.

The human and software diagrams must therefore remain visibly different



The question marks are not gaps that a `.qbin` checksum can fill. They represent unresolved bridge principles.

Whole-brain emulation and the limits of current measurement

What the field has actually achieved

Sandberg and Bostrom's 2008 *Whole Brain Emulation: A Roadmap* remains historically important because it separates acquisition, model construction and simulation rather than treating uploading as one operation. [14] A 2025 *State of Brain Emulation* technical report updates that discussion, but it is a technical synthesis rather than evidence that human WBE has been achieved. [36] The primary experimental literature gives a clearer picture of the actual state of the field.

Capability	Representative frontier	What has been demonstrated	Major missing element for WBE
Whole-brain synaptic wiring in a complex animal	Adult *Drosophila* connectome, Nature 2024	About 139,000 neurons and tens of millions of synapses reconstructed at whole-brain scale	Detailed mammalian/human scale; dynamic and molecular state [15]

Capability	Representative frontier	What has been demonstrated	Major missing element for WBE
Nanoscale human cortical structure	Shapson-Coe et al., Science 2024	~1 mm^3 human temporal cortex; ~57,000 cells, ~150 million synapses, ~1.4 PB reconstructed dataset	It is a minute fraction of a human brain and largely structural [16]
Functional + structural connectomics	MICrONS, Nature 2025	Functional measurements from roughly 75,000 neurons co-registered with a dense EM reconstruction containing >200,000 cells and ~0.5 billion synapses in mouse visual cortex	Whole brain; exhaustive dynamic state; longitudinal molecular variables [37]
Brain-wide electrophysiology	International Brain Laboratory, Nature 2025	Hundreds of thousands of recorded units across hundreds of probes, many mice and brain regions	Not simultaneous exhaustive recording of every neuron/ synapse in one brain [38]
Whole-brain multimodal cellular mapping	WARP zebrafish preprint, 2026	Brain-wide activity during behavior registered with cellular gene-expression/ anatomical information	Small vertebrate; preprint; not complete molecular/ dynamical state or emulation [39]
Human multiscale structural bridging	2026 ex-vivo hemisphere preprint	Links diffusion MRI, X-ray methods and EM across scales, resolving whole-hemisphere pathways down toward individual axons in sampled regions	Not synapse-complete whole-human connectome; ex vivo; no functional state [40]
Whole-neuron cell-type projectomics	Cell Reports 2026	Genetic identity plus complete neuronal morphologies across mouse cortical areas reveals cell-type-specific long-range wiring principles	Still sparse sampling relative to every neuron/synapse and no full dynamical state [41]

The trajectory is impressive but does not support the proposition that current technology can capture a complete human computationally relevant brain state. Indeed, the 2026 literature reinforces rather than eliminates the central problem: increasingly sophisticated studies combine structure, molecular identity, and activity precisely because no single modality supplies them all. [42]

A structural connectome answers approximately:

who is connected to whom, and how?

It does not automatically specify at one instant:

$$\begin{aligned}&\text{membrane electrical state},\\&\text{synaptic efficacy and short-term plasticity},\\&\text{channel/receptor state},\\&\text{neuromodulator fields},\\&\text{intracellular signaling},\\&\text{gene/protein expression},\\&\text{glial contribution},\\&\text{ongoing plasticity},\\&\text{peripheral/body coupling},\\&\text{developmental/history-dependent variables}.\end{aligned}$$

Whether every item in that conceptual list is required for a functionally adequate emulation is itself an open empirical question. The crucial point is that **connectome sufficiency is a hypothesis, not an established result**. Recent multimodal work that explicitly joins cellular activity to gene expression illustrates why biological state cannot simply be assumed to reduce to a connectivity adjacency matrix. [43]

Thus the WBE problem has at least four distinct closure conditions:

$$\text{acquisition sufficiency} \wedge \text{model sufficiency} \wedge \text{simulation fidelity} \wedge \text{validation adequacy}.$$

Even if those were eventually satisfied for function, an additional philosophical premise would still be required to conclude numerical or phenomenal identity.

Brillouin microscopy: what it measures

Brillouin microscopy is a real and increasingly capable biomechanical imaging modality. It detects frequency shifts caused by light interacting with acoustic phonons in matter. In a standard geometry, the Brillouin frequency shift depends on quantities including refractive index, density and the material's high-frequency longitudinal modulus. [44]

A schematic relation is:

$$\omega_B \propto \frac{n}{\lambda}\sqrt{\frac{M}{\rho}},$$

with geometry-dependent factors, where M is the longitudinal modulus and ρ density. [45]

Calling it simply a "stiffness camera" is misleading. Work on hydrated materials showed that water content can strongly dominate the measured shift and generated an important debate over the relationship between Brillouin observables and low-frequency quantities such as Young's modulus; later reviews consequently emphasize careful interpretation of the optical/mechanical quantities. [46]

The technology has advanced substantially. Modern full-field and stimulated/gain implementations have dramatically increased acquisition speed, and 2025–2026 studies demonstrate real-time or high-throughput measurements in living biological samples and mechanically relevant measurements in tissues, including brain-related applications. [47]

Those achievements do **not** change the observable. Brillouin imaging is reporting local material/viscoelastic information through light–acoustic interactions; it is not directly reading autobiographical memory, semantic content, a personal-identity token, or first-person awareness.

No established scientific evidence located in this review connects Brillouin microscopy measurements to personal identity or phenomenal continuity.

That sentence should replace any historical Nexus suggestion that an optical “resonance signature” could certify preservation of a person.

NV-diamond magnetometry: what it measures

Nitrogen-vacancy centers in diamond are quantum spin defects whose energy levels depend on the local magnetic field and can be interrogated optically, including at room temperature. The relevant observable is therefore a **magnetic field**, not cognition or identity.

Barry and colleagues demonstrated optical magnetic detection of single-neuron action potentials using NV centers in diamond under ambient conditions, including excised neuronal preparations and measurements involving intact marine invertebrates. [19] This is important: magnetic fields generated by electrophysiological activity are physically measurable without electrodes contacting the neuron.

But the chain

$$\text{magnetic-field detection} \rightarrow \text{neural-activity detection} \rightarrow \text{cognitive-state reconstruction} \rightarrow \text{identity detection}$$

contains several additional inference problems. Only the early part has been demonstrated.

Recent NV work has improved multiplexing and nanoscale sensing, while sample–sensor stand-off, sensitivity, geometry, optical access and scaling remain important constraints; impressive nanometer-scale sensor capabilities in materials experiments do not translate automatically into simultaneous whole-brain neuronal readout. [48] A 2026 demonstration of transient electrical sensing under bias-field-free conditions used controlled biomimetic electrical signals; it is not a demonstration of whole-neural-state decoding. [49]

Again:

No established scientific evidence located in this review demonstrates an identity-, memory-content-, or phenomenal-continuity-specific magnetic signature measurable by NV-diamond magnetometry.

The two sensor modalities compared

Dimension	Brillouin microscopy	NV-diamond magnetometry
Direct physical observable	Brillouin frequency shift from photon–phonon interaction	Magnetic field via NV spin resonance
Most direct biological interpretation	High-frequency longitudinal mechanical/viscoelastic properties, affected by optical/material variables	Magnetic consequences of electrical/ current activity
Spatial regime	Optical-microscopy scale in accessible samples; implementation-dependent	From nanoscale sensor physics to micrometer-scale biological arrangements; effective resolution strongly limited by stand-off and geometry
Temporal regime	Historically spectroscopy-limited; recent systems greatly accelerate acquisition toward real-time/high-throughput biological imaging	Can resolve transient signals including action potentials in demonstrated preparations
Biomedical maturity	Cells, tissues, developmental biology, biomechanics; expanding in vivo applications	Powerful quantum sensing platform; neuronal detection remains much less mature and harder to scale
Deep-brain non-invasive whole-human use	Not demonstrated	Not demonstrated
Measures memory contents?	No evidence	No evidence
Measures personal identity?	No evidence	No evidence
Measures phenomenal continuity?	No evidence	No evidence
Potential relevance to a future emulation program	Could contribute local material-state constraints	Could contribute electrophysiological-field constraints
Legitimate Nexus classification	Instrument modality, not identity meter	Instrument modality, not identity meter

The measurement literature supporting these distinctions is direct. [50]

QHRF: a schema is not a scientific observable

The internal QHRF work presents a particularly useful example of implementation status being confused with scientific status.

Nexus has an NWB example that can write and read QHRF-specific extension types. The implementation includes a processing module, a `QHRFContainer`, time series, a modes table, optional frequency/phase fields and smoke tests covering round-trip serialization. [8]

That means:

$$\boxed{\text{QHRF data representation: IMPLEMENTED / testable}}$$

It does **not** mean:

$$\boxed{\text{QHRF as identity observable: VALIDATED}}$$

The independent literature search conducted for this report did not locate a peer-reviewed measurement theory that defines QHRF as a personal-identity or phenomenal-continuity observable, supplies a calibrated forward model from an identity state to Brillouin/NV or other sensor outputs, and independently validates predictions from that model.

Accordingly, the scientifically correct present classification is:

> **QHRF, insofar as it is proposed as a latent identity/phenomenal-continuity variable, is a HISTORICAL / SPECULATIVE CONSTRUCT. Its NWB serialization infrastructure can nevertheless be a real implemented software artifact.**

The Bayesian issue is straightforward. Suppose H is "identity preserved" and Y is a collection of measurements. Bayes's rule says:

$$P(H \mid Y) = \frac{p(Y \mid H)P(H)}{p(Y)}.$$

Writing the equation is not enough. The scientifically substantive term is the likelihood:

$$p(Y \mid H).$$

A posterior about identity is empirically meaningful only if there is a justified model relating the hypothesized identity condition to the distribution of observations and if competing latent explanations are sufficiently identifiable. Statistical work on Bayesian modeling and identifiability emphasizes that priors are interpreted through the likelihood and that poorly identified models can leave posterior conclusions highly dependent on prior assumptions. [9]

Consequently:

$$\text{Brillouin data} + \text{NV data} + \text{Bayes} \not\Rightarrow P(\text{personal identity preserved})$$

unless a validated scientific bridge exists between those observables and the identity hypothesis.

A defensible QHRF research claim would first need all of the following conceptually: a clearly dimensioned latent quantity; a forward observation model; a measurement/noise likelihood; calibration standards; demonstrated discriminant or predictive validity; identifiability analysis; prospective falsifiable predictions; and independent replication. Until then, posterior percentages would quantify assumptions **inside the stipulated model**, not empirical identification of phenomenal continuity.

Boundaries, adversarial review, claim audit, and research program

What QMT can legitimately claim

The strongest defensible Nexus result is narrower than "identity transfer" but more substantial than ordinary backup/restore:

> **Given a formally specified digital entity, an explicit continuity contract, a versioned state semantics, a sufficiently closed dependency set, a recorded transformation trajectory, and explicit branch rules, QMT can in principle verify whether a candidate destination is an admissible continuation of the source's architectural lineage.**

This claim is credible because each constituent obligation has established analogues: abstraction and refinement mappings in formal methods; observational equivalence and trace semantics for behavior; deterministic replay and checkpoint/restart for reconstructing execution state; provenance records for historical derivation; cryptographic commitments for integrity; and authentication/authorization standards for separating identity claims from permissions. [51]

For current Nexus specifically, a weaker statement is already supported: the 2026 reference implementation demonstrated deterministic state-representation and restoration properties in its stated test environment, including byte-identical seeded snapshot reproduction and restoration of selected runtime state. [2]

Thus a carefully worded canonical claim could be:

> **Nexus implements tested computational state-preservation mechanisms. QMT is a proposed formal continuity layer that should determine lineage continuity through explicit proof obligations over state, invariants, behavior, provenance, dependency closure and branch semantics.**

What QMT cannot legitimately claim

Nothing in the examined Nexus test evidence or the external literature establishes the following implications:

$$\text{snapshot restored} \Rightarrow \text{consciousness restored}$$

$$\text{deterministic computation} \Rightarrow \text{same human person}$$

$$\text{behaviorally indistinguishable} \Rightarrow \text{numerically identical}$$

$$\text{same architectural lineage} \Rightarrow \text{same first-person subject}$$

$$\text{sensor signature reproduced} \Rightarrow \text{phenomenal continuity}.$$

The first implication is explicitly rejected by the Nexus verification report itself. [2] The remaining implications are disputed by the uploading and personal-identity literature and lack a demonstrated consciousness-continuity measurement bridge. [52]

It follows that "identity transfer" should not be the scientific label for Nexus's demonstrated capability. **Computational continuity**, **state migration**, **lineage-preserving reconstruction**, or **formal continuity verification** are defensible labels. Human personal-identity transfer remains a philosophical hypothesis layered on top of those concepts.

Claim-status audit

"Canonical" below means "part of Nexus's own current or historical architecture," not "scientifically proven."

Nexus-adjacent claim	Status	Audit judgment
Quinary packed-state round trips work on tested cases	**VERIFIED**	Supported by Nexus v4 property tests [2]
Tested runtime principal state can be saved and restored	**VERIFIED**	Supported under recorded implementation/environment [2]
Seeded demonstration `.qbin` can be reproduced byte-for-byte in the tested environment	**VERIFIED**	Strong but environment-bounded result [2]
`.qbin` is already demonstrated to be cross-platform byte-identical	**NOT VERIFIED / DESIGN TARGET**	Nexus verification explicitly lists cross-platform determinism as unestablished [2]
TMS is the intended live state continuum	**CANONICAL / ARCHITECTURAL**	Internal Nexus design role, not externally validated performance claim [4]
HDF5 is intended as durable persistence	**CANONICAL / ARCHITECTURAL**	Technically consistent with HDF5's general storage capabilities; Nexus long-horizon design remains to be validated [4] [53]
`.qbin` should be a versioned computational continuity package	**PROPOSED / DESIGN TARGET**	Strongly defensible rewrite of historical state-archive concept
QMT can combine invariants, provenance, state equivalence, behavioral conformance and fork rules	**PROPOSED / FORMALLY SPECIFIABLE**	Grounded in established formal concepts; requires an executable specification to become verified [54]

Nexus-adjacent claim	Status	Audit judgment
A QMT pass establishes architectural software lineage under its contract	**DESIGN TARGET**	Defensible if semantics are explicit and verifier sound
A QMT pass establishes human personal identity	**RESEARCH HYPOTHESIS / NOT ESTABLISHED**	Philosophy remains divided
A QMT pass establishes phenomenal continuity	**NOT ESTABLISHED**	No accepted bridge principle or external measurement
A checksum match proves "identity remains unbroken"	**HISTORICAL/ UNSUPPORTED INTERPRETATION**	Hashes establish integrity/equality of committed data, not consciousness [4]
"Qualia Packets" provide a scientific continuity measurement	**HISTORICAL / SPECULATIVE**	No independent validation located
`.qbin` "serializes consciousness"	**HISTORICAL/ SUPERSEDED**	Conflates computational checkpointing with a philosophical/phenomenal conclusion [3]
QHRF NWB objects can be serialized and round-tripped	**IMPLEMENTED / INTERNALLY VERIFIED**	Supported by the QHRF example and smoke-test design [8]
QHRF is a validated identity/ phenomenal-continuity observable	**HISTORICAL / SPECULATIVE CONSTRUCT**	No validated independent measurement model located
Brillouin microscopy measures a unique identity resonance	**NOT ESTABLISHED**	Literature supports mechanical/ viscoelastic observables instead [55]
NV magnetometry measures a unique identity or memory signature	**NOT ESTABLISHED**	Literature supports magnetic-field/ neural-signal sensing instead [19]
Current WBE technology can emulate a human brain at identity-relevant fidelity	**NOT DEMONSTRATED**	Frontier data remain many levels short of this claim [56]

Adversarial review: how the computational thesis could fail

The central digital-continuity thesis should itself be falsifiable. The following findings would materially weaken it.

Hidden-state failure. If two reconstructions satisfy `.qbin` completeness and QMT state checks yet systematically diverge under identical declared inputs because consequential state was omitted, then the

continuity contract's closure specification is false. Replay work makes clear why omitted nondeterminism is a real risk. [12]

Semantic migration failure. If version transformations preserve byte-level fields but alter their meaning—for example through numerical precision, changed HTNLA operators, altered model weights or schema semantics—then representation migration is not state continuity. QMT must test the abstraction relation, not just data presence.

Observation-domain failure. If a destination passes behavioral equivalence only because the test suite probes a narrow domain while materially relevant behavior has changed elsewhere, then the behavioral claim must be narrowed. Formal equivalence is always relative to the transition systems and observations supplied to it. [10]

Fork incoherence. If Nexus allows multiple simultaneously authoritative descendants to claim an exclusive one-to-one identity while retaining no branch semantics, its identity relation becomes logically and operationally incoherent. Parfit's fission problem and distributed split-brain engineering converge on this point for different reasons. [57]

Provenance failure. If a fresh copy can manufacture the same identity metadata without an authenticated causal/provenance connection, QMT has reduced identity to self-asserted labels. Integrity and signatures must therefore bind provenance evidence, while external authorization remains independent.

Substrate-independence failure. If restoration works only on one exact Python/NumPy/CPU environment and semantic state cannot be preserved under declared cross-platform transformations, the broader substrate-independence claim must be restricted. Nexus's own verification report currently acknowledges this exact gap. [2]

Constitutional insufficiency. Even a perfectly implemented QMT can produce misleadingly strong continuity claims if its invariant set omits properties stakeholders actually mean by "the entity." Formal verification proves a system satisfies its specification; it does not prove that the specification captured the right philosophical concept.

For the human extension, the thesis would be materially weakened by strong evidence that cognition or consciousness depends essentially on biological variables not preserved by the proposed abstraction—for example, substrate-specific processes whose replacement produces systematic loss of relevant function despite preservation of computational organization. Conversely, finding ever-stronger organizational invariance across radically different implementations would strengthen functionalist premises, but would still require argument to reach strict numerical or phenomenal identity.

What could ever be measured?

Property	Current measurability	Conceivable empirical test	Formal criterion	Residual issue
Serialized bits	Excellent	Hash/checksum	Exact equality	No semantic conclusion

Property	Current measurability	Conceivable empirical test	Formal criterion	Residual issue
Logical state	Excellent for engineered systems	Decode/ canonicalize and compare	State relation R_S	Depends on adequate schema
Replay fidelity	Good in bounded engineered systems	Re-execute controlled traces	Trace equivalence/ tolerance	Environment/ nondeterminism closure
Behavioral continuity	Good within specified domains	Conformance/ behavior suites	Bisimulation, trace inclusion, refinement	Domain completeness
Software lineage	Good if architecture defines it	Provenance and generation verification	QMT lineage predicate	Conventional rather than metaphysical
Human memories/ dispositions	Partial	Cognitive/ behavioral/neural tests	No universally sufficient formal set	Theory of psychological identity
Whole-brain functional equivalence	Not currently achievable at full human scale	Hypothetical multimodal longitudinal comparison	Could eventually be specified	Which biological detail is sufficient?
Consciousness presence	Correlates are measurable	Theory-dependent behavioral/neural paradigms	No accepted theory-neutral certificate	Hard problem/ theory disagreement
Numerical human identity	No accepted meter	Evidence can constrain theories	Theory-dependent	May not be an empirical natural-kind property
First-person phenomenal continuity	No accepted external continuity measurement	Unknown; depends on theory of consciousness	None presently accepted	Potentially irreducible epistemic/ metaphysical gap

Open research questions and recommended next steps

The next Nexus continuity paper should not pursue a human-transfer protocol. It should pursue **formal precision**.

First, QMT needs an explicit mathematical semantics. The highest-value work is to define the entity state space, abstraction/canonicalization functions, invariants, permissible transformation relation, observation alphabet, provenance relation and branch semantics, then determine which obligations can be model-

checked, proved, replayed or merely tested. Refinement mappings and observational equivalence provide mature starting points. [54]

Second, `.qbin` needs a published versioned schema with canonical encodings and migration semantics. The current snapshot proves a useful implementation baseline, but Nexus itself acknowledges that cross-platform reproducibility has not been established. [2] Cross-version/cross-runtime test vectors should therefore become more important than another same-environment byte hash.

Third, continuity needs explicit **branch algebra**. Nexus should state whether one source can have multiple valid descendants, whether "identity" refers to a lineage or a currently authoritative token, and what happens at merge, replay, rollback and fork. A provenance DAG is probably a more faithful mathematical object than a simple equivalence class once branching is allowed.

Fourth, QMT should adopt at least three result states: `PASS`, `FAIL`, and `UNRESOLVED`. Unknown dependencies, unavailable external state, unsupported schema migrations or unverified numerical transformations must not be silently coerced into either continuity or discontinuity.

Fifth, authentication and authorization should be formally excluded from the identity predicate. Continuity metadata can identify an architectural lineage; credentials and capabilities should be revalidated under independent policy. Existing authorization standards illustrate why this separation is mature security practice rather than a philosophical nicety. [25]

Sixth, the historical QHRF/phenomenal-continuity material should be rewritten as an epistemic case study: it shows how an implemented data schema and sophisticated Bayesian vocabulary can outpace scientific identification of the latent variable they purport to represent. That is a valuable methodological lesson, not merely material to discard.

Finally, the human portion should remain explicitly conditional:

$$\text{If theory } M \text{ says property set } P \text{ is sufficient for consciousness/personhood,}$$

then future science might test preservation of P.

But Nexus should never silently replace that conditional with:

$$\text{QMT passed} \therefore \text{person survived.}$$

That is precisely the boundary this investigation establishes.

Selected primary and authoritative references

The citations throughout this report link directly to the inspected source records. The following are the load-bearing references rather than an indiscriminate bibliography.

Reference	Role in this analysis
Parfit, Derek. *Reasons and Persons*. Oxford University Press, 1984.	Numerical identity, psychological continuity, fission and "what matters." [29]
Chalmers, David J. "The Singularity: A Philosophical Analysis." *Journal of Consciousness Studies* 17, 2010.	Uploading, consciousness and philosophical consequences of emulation. [58]
Chalmers, David J. "Uploading: A Philosophical Analysis." 2014.	Destructive, gradual and nondestructive uploading; consciousness and identity distinctions. [59]
Corabi, Joseph, and Susan Schneider. "If You Upload, Will You Survive?" 2014.	Skeptical analysis of upload survival. [60]
Wiley, Keith B., and Randal A. Koene. "The Fallacy of Favoring Gradual Replacement Mind Uploading Over Scan-and-Copy."	Challenges metaphysical privilege assigned to gradual replacement. [61]
Hennessy, Matthew, and Robin Milner. "Algebraic Laws for Nondeterminism and Concurrency." *Journal of the ACM* 32(1), 1985, 137–161. doi:10.1145/2455.2460.	Observational equivalence and process semantics. [10]
Abadi, Martín, and Leslie Lamport. "The Existence of Refinement Mappings." *Theoretical Computer Science* 82(2), 1991, 253–284. doi:10.1016/0304-3975(91)90224-P.	Implementation-to-specification refinement. [11]
Dunlap, George W., et al. "ReVirt: Enabling Intrusion Analysis through Virtual-Machine Logging and Replay." OSDI, 2002.	Capturing nondeterministic events for deterministic replay. [12]
Ansel, Jason, Kapil Arya, and Gene Cooperman. DMTCP checkpointing work, 2007.	Distributed checkpoint/restart and broader state closure. [13]
Sandberg, Anders, and Nick Bostrom. *Whole Brain Emulation: A Roadmap*. Future of Humanity Institute, Oxford University, 2008.	Foundational WBE decomposition. [14]
Dorkenwald, Sven, et al. "Neuronal Wiring Diagram of an Adult Brain." *Nature* 634, 2024.	Whole adult *Drosophila* connectome. [15]
Shapson-Coe, Alexander, et al. "A Petavoxel Fragment of Human Cerebral Cortex Reconstructed at Nanoscale Resolution." *Science*, 2024. doi:10.1126/science.adk4858.	Human nanoscale cortical reconstruction and data-scale benchmark. [16]
MICrONS Consortium. "Functional Connectomics Spanning Multiple Areas of Mouse Visual Cortex." *Nature*, 2025.	Joint functional and dense structural mapping. [37]

Reference	Role in this analysis
International Brain Laboratory. "A Brain-Wide Map of Neural Activity During Complex Behaviour." *Nature*, 2025.	Large multisite brain-wide electrophysiology benchmark. [38]
Cogitate Consortium. "Adversarial Testing of Global Neuronal Workspace and Integrated Information Theories of Consciousness." *Nature* 642, 2025, 133–142. doi:10.1038/s41586-025-08888-1.	Current evidence that consciousness science remains theoretically contested. [17]
Scarcelli, Giuliano, et al. Brillouin microscopy work, *Nature Methods*, 2015.	Biological Brillouin mechanical imaging. [62]
Wu et al. Brillouin/hydration analysis, 2018.	Demonstrates why Brillouin shift cannot be naively equated to conventional stiffness. [45]
Barry, John F., et al. "Optical Magnetic Detection of Single-Neuron Action Potentials Using Quantum Defects in Diamond." *PNAS* 113, 2016. doi:10.1073/pnas.1601513113.	Primary demonstration of NV-based neuronal magnetic sensing. [19]
Marquez-Legorreta, Emmanuel, et al. "Whole-Brain Co-Mapping of Gene Expression and Neuronal Activity at Cellular Resolution in Behaving Zebrafish." bioRxiv, 2026. doi:10.64898/2026.02.07.704095.	Recent multimodal whole-brain cellular mapping frontier; preprint status. [43]
W3C PROV family of recommendations.	Standard conceptual machinery for entities, activities, agents and derivational provenance. [63]
IETF RFC 6749; RFC 7628; RFC 8705.	Separation of authorization, authenticated principal information and possession-bound capabilities. [25]
The HDF Group, HDF5 Data Model and datatype documentation.	Durable typed hierarchical storage; illustrates why storage format does not itself define application identity semantics. [23]
Project Nexus v4.0 Verification Report, July 2026.	Primary internal evidence for current tested persistence and deterministic-state claims—and their explicit limitations. [2]
HVD / Nexus 4 Recursive Cognitive Database Architecture: Formal Briefing, 2026.	Internal evidence for TMS/HVD/HDF5 responsibility assignments and for phenomenal claims requiring downgrade. [4]

Reference	Role in this analysis
Historical `.qbin` format notes, March 2026.	Historical evidence of useful checkpoint-state thinking coupled to an unsupported consciousness-serialization inference. [3]
Nexus QHRF NWB Example, 2026.	Establishes implemented data-schema/round-trip infrastructure, not scientific validity of QHRF as an identity observable. [8]

The final boundary can therefore be stated with precision:

> QMT can become a rigorous verifier of
> specified computational continuity and lineage.

but

> Neither QMT, ‘.qbin‘, deterministic replay,
> Brillouin microscopy, NV magnetometry, nor QHRF
> currently establishes human numerical or phenomenal continuity.

That is not a weakness of the formal-continuity program. It is its most important clarification. By restricting QMT to claims for which proof obligations can actually be stated and evidence can actually be produced, Nexus can turn an historically speculative “identity transfer” narrative into a rigorous formal-methods contribution: **the verification of digital continuity across transformations, together with an explicit proof boundary showing exactly where software engineering ends and unresolved philosophy of mind begins.**

---

[1] [11] [51] [54] https://www.sciencedirect.com/science/article/pii/030439759190224P
https://www.sciencedirect.com/science/article/pii/030439759190224P

[2] PROJECT_NEXUS_V4_VERIFICATION_REPORT.md
https://drive.google.com/file/d/1FxHnbRAKf1QNEHjO5UUnSBI4XYMcleXL/view?usp=drivesdk

[3] consciousness.qbun file format notes
https://docs.google.com/document/d/1fjKqv9La9rRpfUy1gP6_ZFMKnfgQKBFKpgwuAcfK2CA/edit?usp=drivesdk

[4] HVD / Nexus 4 Recursive Cognitive Database Architecture: Formal Briefing
https://docs.google.com/document/d/1AteSGSm49kTUZR6wqn6NUP86BYSSeLc2sRIKp9-JkhY/edit?usp=drivesdk

[5] [26] [27] [52] [57] https://academic.oup.com/book/12484/chapter-abstract/163166818
https://academic.oup.com/book/12484/chapter-abstract/163166818

[6] [15] https://www.nature.com/articles/s41586-024-07558-y
https://www.nature.com/articles/s41586-024-07558-y

7 45 46 55 https://www.nature.com/articles/s41592-018-0076-1
https://www.nature.com/articles/s41592-018-0076-1

8 README.md
https://drive.google.com/file/d/1eAMypmbiAVg7fF2nslddvKObbDXaos1C/view?usp=drivesdk

9 https://arxiv.org/abs/1708.07487
https://arxiv.org/abs/1708.07487

10 22 https://www.research.ed.ac.uk/en/publications/algebraic-laws-for-nondeterminism-and-concurrency/
https://www.research.ed.ac.uk/en/publications/algebraic-laws-for-nondeterminism-and-concurrency/

12 24 https://www.usenix.org/conference/osdi-02/revirt-enabling-intrusion-analysis-through-virtual-machine-logging-and-replay
https://www.usenix.org/conference/osdi-02/revirt-enabling-intrusion-analysis-through-virtual-machine-logging-and-replay

13 https://arxiv.org/abs/cs/0701037
https://arxiv.org/abs/cs/0701037

14 https://ora.ox.ac.uk/bookmarks/uuid%3Aa6880196-34c7-47a0-80f1-74d32ab98788
https://ora.ox.ac.uk/bookmarks/uuid%3Aa6880196-34c7-47a0-80f1-74d32ab98788

16 56 https://pubmed.ncbi.nlm.nih.gov/38723085/
https://pubmed.ncbi.nlm.nih.gov/38723085/

17 https://www.nature.com/articles/s41586-025-08888-1
https://www.nature.com/articles/s41586-025-08888-1

18 44 50 62 https://www.nature.com/articles/nmeth.3616
https://www.nature.com/articles/nmeth.3616

19 https://pubmed.ncbi.nlm.nih.gov/27911765/
https://pubmed.ncbi.nlm.nih.gov/27911765/

20 29 https://academic.oup.com/book/12484/chapter-abstract/163165934
https://academic.oup.com/book/12484/chapter-abstract/163165934

21 https://www.usenix.org/conference/osdi-02/design-and-implementation-zap-system-migrating-computing-environments
https://www.usenix.org/conference/osdi-02/design-and-implementation-zap-system-migrating-computing-environments

23 53 https://portal.hdfgroup.org/documentation/hdf5/latest/_h5_d_m__u_g.html
https://portal.hdfgroup.org/documentation/hdf5/latest/_h5_d_m__u_g.html

25 https://www.rfc-editor.org/info/rfc6749/
https://www.rfc-editor.org/info/rfc6749/

28 34 59 https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118736302.ch6
https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118736302.ch6

30 https://philpapers.org/rec/DANTOP
https://philpapers.org/rec/DANTOP

31 32 https://survey2020.philpeople.org/survey/results/all
https://survey2020.philpeople.org/survey/results/all

33 https://survey2020.philpeople.org/survey/results/4914
https://survey2020.philpeople.org/survey/results/4914

35 60 https://onlinelibrary.wiley.com/doi/10.1002/9781118736302.ch8
https://onlinelibrary.wiley.com/doi/10.1002/9781118736302.ch8

36 https://arxiv.org/abs/2510.15745
https://arxiv.org/abs/2510.15745

37 https://www.nature.com/articles/s41586-025-08790-w
https://www.nature.com/articles/s41586-025-08790-w

38 https://www.nature.com/articles/s41586-025-09235-0
https://www.nature.com/articles/s41586-025-09235-0

39 https://www.janelia.org/publication/whole-brain-co-mapping-of-gene-expression-and-neuronal-activity-at-cellular-resolution
https://www.janelia.org/publication/whole-brain-co-mapping-of-gene-expression-and-neuronal-activity-at-cellular-resolution

40 https://sensein.group/publications/10.64898/2026.04.02.716198/
https://sensein.group/publications/10.64898/2026.04.02.716198/

41 https://pubmed.ncbi.nlm.nih.gov/42301808/
https://pubmed.ncbi.nlm.nih.gov/42301808/

42 43 https://www.biorxiv.org/content/10.64898/2026.02.07.704095v1
https://www.biorxiv.org/content/10.64898/2026.02.07.704095v1

47 https://www.nature.com/articles/s41566-025-01619-y
https://www.nature.com/articles/s41566-025-01619-y

48 https://journals.aps.org/prx/abstract/10.1103/t8fz-3tzs
https://journals.aps.org/prx/abstract/10.1103/t8fz-3tzs

49 https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202500967
https://advanced.onlinelibrary.wiley.com/doi/10.1002/admi.202500967

58 https://openresearch-repository.anu.edu.au/items/ac84d4a1-5cbc-4c6a-ac9e-704ec95c9b9f
https://openresearch-repository.anu.edu.au/items/ac84d4a1-5cbc-4c6a-ac9e-704ec95c9b9f

61 https://arxiv.org/abs/1504.06320
https://arxiv.org/abs/1504.06320

63 https://www.w3.org/TR/2013/REC-prov-n-20130430/
https://www.w3.org/TR/2013/REC-prov-n-20130430/